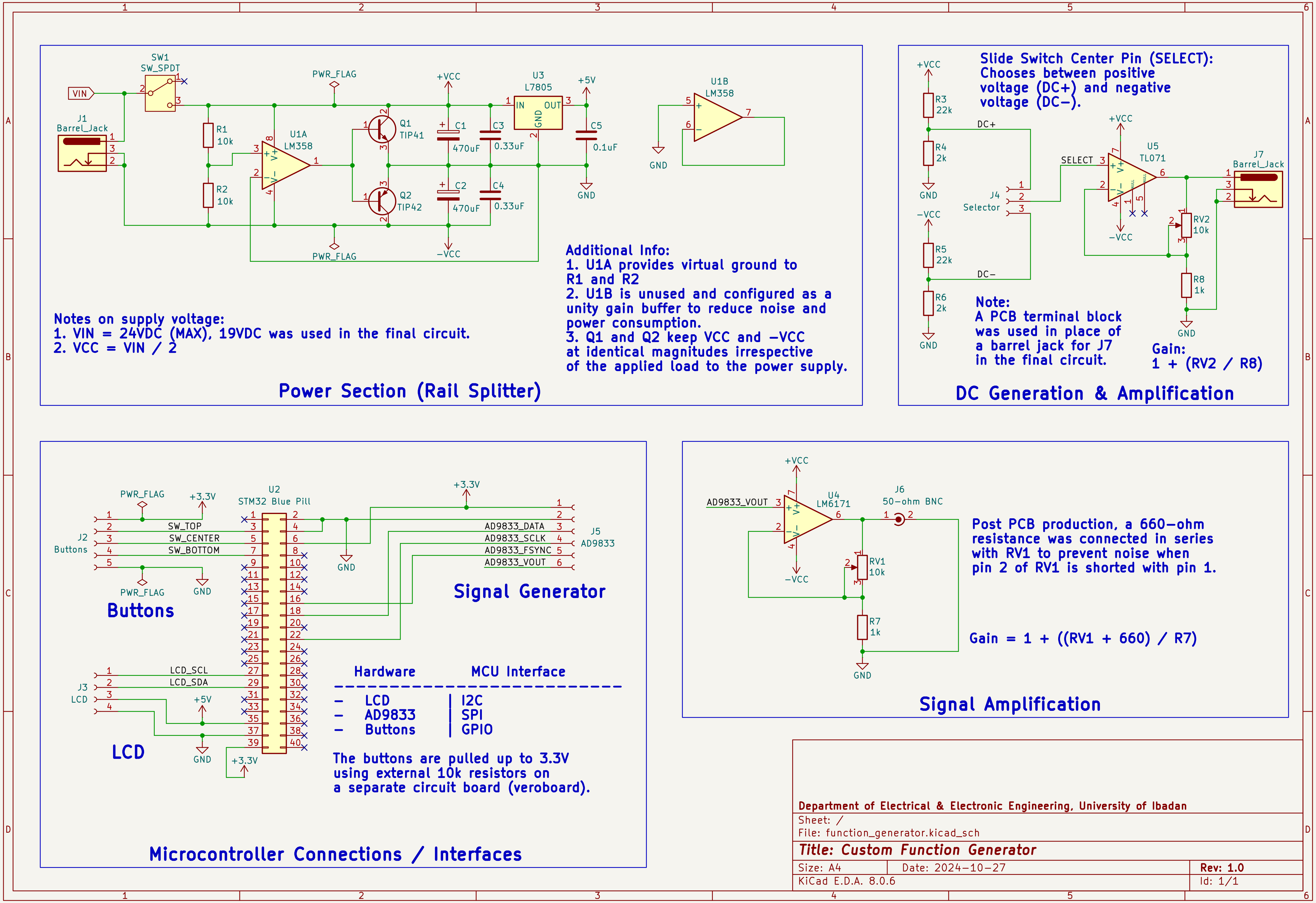
Power Section (Rail Splitter)
SW1
SW_SPDT
VIN
J1
Barrel_Jack
PWR_FLAG
+VCC
U3
L7805
IN
OUT
GND
+5V
R1
10k
R2
10k
U1A
LM358
Q1
TIP41
Q2
TIP42
C1
470uF
C2
470uF
C3
0.33uF
C4
0.33uF
C5
0.1uF
GND
-VCC
U1B
LM358
Notes on supply voltage:
1. VIN = 24VDC (MAX). 19VDC was used in the final circuit.
2. VCC = VIN / 2
Additional Info:
1. U1A provides virtual ground to R1 and R2
2. U1B is unused and configured as a unity gain buffer to reduce noise and power consumption.
3. Q1 and Q2 keep VCC and -VCC at identical magnitudes irrespective of the applied load to the power supply.
DC Generation & Amplification
Slide Switch Center Pin (SELECT): Chooses between positive voltage (DC+) and negative voltage (DC-).
R3
22k
R4
2k
R5
22k
R6
2k
DC+
DC-
J4
Selector
SELECT
U5
TL071
RV2
10k
R8
1k
J7
Barrel_Jack
Note:
A PCB terminal block was used in place of a barrel jack for J7 in the final circuit.
Gain:
1 + (RV2 / R8)
Microcontroller Connections / Interfaces
J2
Buttons
SW_TOP
SW_CENTER
SW_BOTTOM
+3.3V
U2
STM32 Blue Pill
J5
AD9833
AD9833_DATA
AD9833_SCLK
AD9833_FSYNC
AD9833_VOUT
Signal Generator
Buttons
J3
LCD
LCD_SCL
LCD_SDA
LCD
Hardware | MCU Interface
- LCD | I2C
- AD9833 | SPI
- Buttons | GPIO
The buttons are pulled up to 3.3V using external 10k resistors on a separate circuit board (veroboard).
Signal Amplification
AD9833_VOUT
U4
LM6171
J6
50-ohm BNC
RV1
10k
R7
1k
Post PCB production, a 660-ohm resistance was connected in series with RV1 to prevent noise when pin 2 of RV1 is shorted with pin 1.
Gain = 1 + ((RV1 + 660) / R7)
Department of Electrical & Electronic Engineering, University of Ibadan
Sheet: /
File: function_generator.kicad_sch
Title: Custom Function Generator
Size: A4
Date: 2024-10-27
Rev: 1.0
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Id: 1/1